

Scalability & Future Plan

Date	29/09/2026
Team ID	SWTID-2026-1297
Project Name	AI FAQ Assistant API
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.N o	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited to a single server instance	Performance may reduce when the number of users increases	High
2	MongoDB storage may grow with increasing FAQ and user data	Larger datasets may require better storage and database optimization	Medium
3	Dependence on Google Gemini API for AI responses	API availability, usage limits, or billing restrictions may affect AI features	High

Scalability Plan:

S.N o	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Runs on a single Express.js server	Deploy using scalable cloud infrastructure and multiple server instances
2	Data Storage	MongoDB database	Use optimized MongoDB indexes and scalable cloud database storage
3	Performance	Suitable for current project-level usage	Add caching, database optimization, and load balancing
4	Security	JWT authentication and bcrypt password hashing	Add HTTPS, stronger validation, rate limiting, and improved access control

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact	
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Phase 2	Improve FAQ search and add advanced filtering	1–2 Months	Faster and more accurate FAQ retrieval	
Phase 3	Deploy the API on cloud infrastructure and improve scalability	3–4 Months	Support more users and requests	
Phase 4	Add analytics, admin dashboard, and enhanced AI features	5–6 Months	Better monitoring, management, and user support	